



Characterizing Chemical Ordering in HEOs by Accurate and Efficient Energy and Force Predictions

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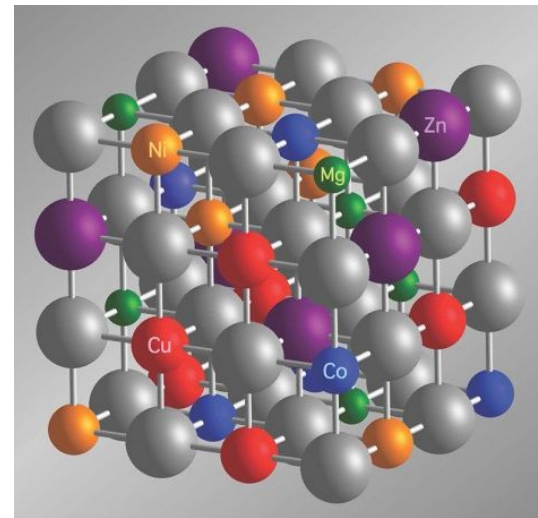
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PROJECT OBJECTIVES

1. The **overall objective** of the project is to chemically characterize high-entropy oxides by efficiently predicting energies and forces using Machine Learning Interatomic Potentials (MLIPs).
2. **Research Questions:**
 - a. Can a machine-learning potential trained on DFT data accurately predict energies and forces in HEOs for reliable geometry optimization?
 - b. Can the trained potential produce physically meaningful phonon spectra to assess the dynamical stability of HEO structures?

HIGH ENTROPY OXIDES (HEOs)

1. HEOs are oxides containing five or more cations in near-equal ratios, all sharing a single crystal lattice.
2. They form stable single-phase structures primarily due to high configurational entropy.
3. Exhibit enhanced functional properties, such as improved ionic conductivity, with applications in catalysis, solid-state batteries, and energy storage.



e.g: $(\text{MgNiCoCuZn})_{0.2}\text{O}$

COMPUTING CONFIGURATIONAL ENTROPY

In this slide, we show the mathematical computation of configurational entropy, and this in turn lowers the free energy and stabilizes single-phase HEO structures.

Many Possible Configurations of A, B, C on N Sites

A	A	A	C	A	B	...		
B	C	B	A	C	A			



$$N = N_A + N_B + N_C$$

$$F = \langle E \rangle - TS_{\text{conf}}$$

$$\Omega = \frac{N!}{N_A! N_B! N_C!}$$

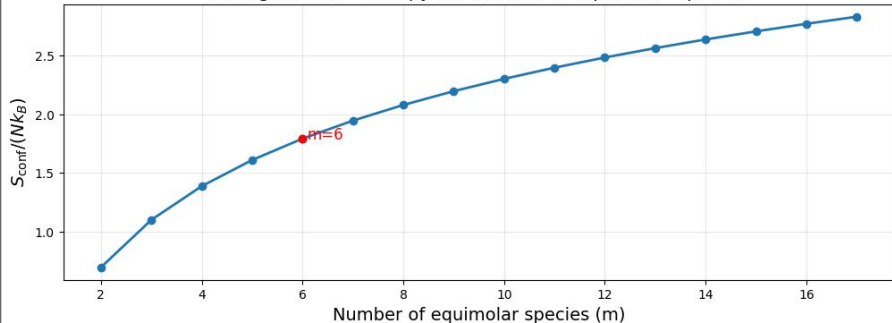
$$S_{\text{conf}} = k_B \ln \Omega = k_B \ln \left(\frac{N!}{N_A! N_B! N_C!} \right)$$

$$\frac{S_{\text{conf}}}{k_B} \approx N \ln N - N_A \ln N_A - N_B \ln N_B - N_C \ln N_C$$

$$x_i = \frac{N_i}{N} \Rightarrow \frac{S_{\text{conf}}}{k_B} = -N \sum_{i=A,B,C} x_i \ln x_i$$

$$S_{\text{conf}} = -Nk_B (x_A \ln x_A + x_B \ln x_B + x_C \ln x_C)$$

Configurational Entropy vs. Number of Equimolar Species



MACHINE-LEARNING INTERATOMIC POTENTIAL FROM DFT

Regular Approaches for Energy Calculation

➤ Classical Force Field Methods.

- Fast & Scalable
- But NOT Accurate

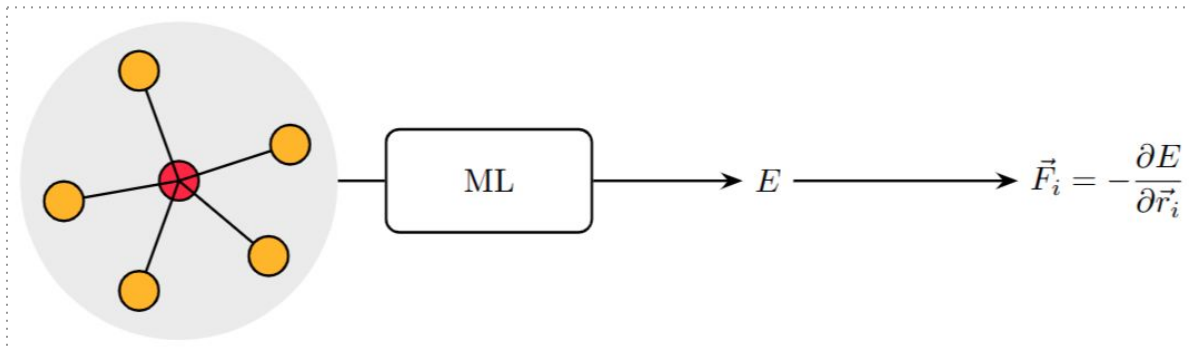
➤ DFT

- Highly Accurate
- But Computationally Expensive ➡ NOT Scalable

These DRAWBACKS makes Energy Calculations in Complex HEOs impractical at Scale

MACHINE-LEARNING INTERATOMIC POTENTIAL FROM DFT

Machine Learning for Energy Calculation



Fit to a limited library of
quantum calculations

Significantly improved speed
and scaling

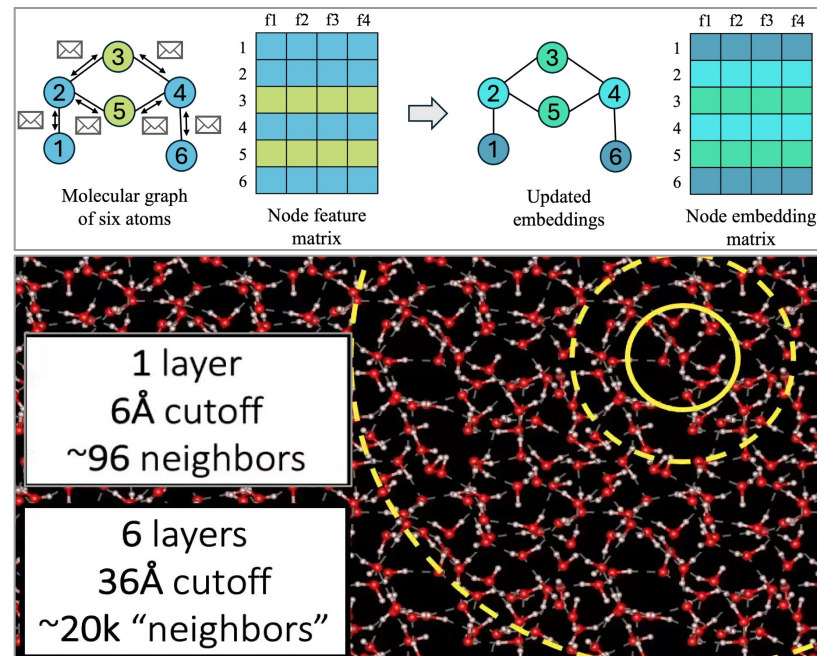
Gives near-quantum accuracy

Classical potential speeds and
scales at ab-initio accuracy.

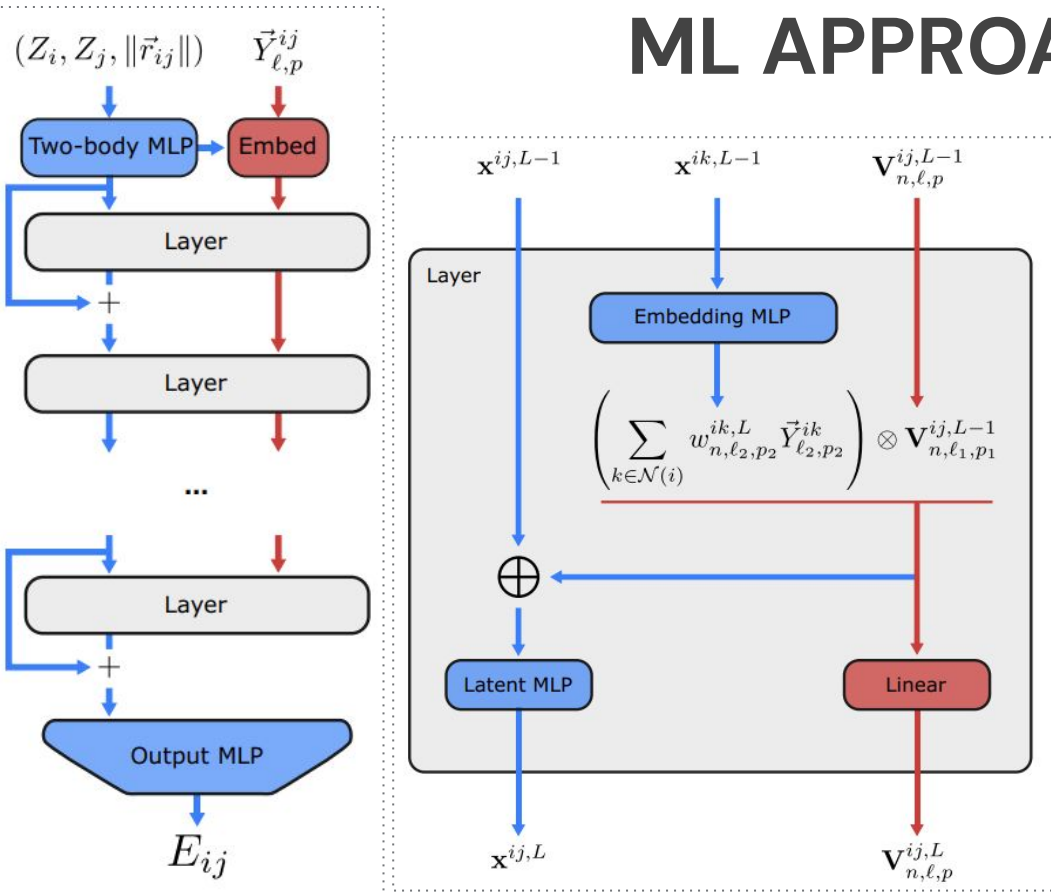
ML APPROACH: LIMITATIONS OF MESSAGE PASSING

Message Passing Neural Networks (MPNNs)

- In this type of NNs, each layer aggregates information from neighboring atoms to build atomic representations.
- Increasing network depth leads to a growing effective interaction cutoff.
- This results in a rapid increase in the number of interacting neighbors and computational cost.
- In HEOs, large supercells and highly diverse local chemical environments make message passing inefficient and difficult to scale.

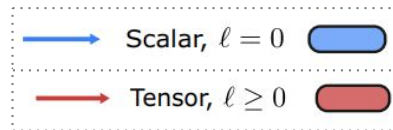


ML APPROACH: ALLEGRO



Allegro Model Architecture

- Allegro is a strictly local, E(3)-equivariant neural network that learns pairwise interactions via tensor products.
- Achieves high accuracy and massive scalability without using message passing.



METHODOLOGY – DATASET PREPARATION

1. We developed a pipeline to transform DFT-generated data into a format compatible with the Allegro model for efficient training and prediction.
2. Our workflow consisted of three main stages: dataset preparation, model architecture and training, and atomistic simulations using ASE.
3. In this study, we constructed the J14Cr dataset comprising atomic structures with varying chemical configurations of six cations: Mg, Co, Ni, Cu, Zn, and Cr.
4. For each configuration, DFT calculations provide the total potential energy & atomic forces, which serve as the regression targets for the model.

*Processed Sample
from the Dataset:*

```
10
Lattice="2.96561 0.0 0.0 0.0 -1.7121957318114465 2.421410425365073 -1.4828050000000008 -7.704880793151508 -7.26423127609522"
Properties=species:S:1:pos:R:3:forces:R:3 energy=-66.35023497 pbc="T T T"
Co      0.00000000      0.00000000      0.00000000      0.19222766      0.29614307      0.08826916
0      1.48280500      -7.70488079      -3.63211564      -0.01444584      0.03488326      0.33583984
Ni     -0.00000000      -6.84878293      -4.84282085      -0.00813510      0.10591445      -0.37992074
0      0.00000000      -5.13658720      -3.63211564      0.14846196      -0.11584893      -0.10519305
Ni     1.48280500      -5.99268506      -2.42141043      0.09666879      -0.16544949      0.04842314
0      1.48280500      -4.28048933      -1.21070521      -0.40930622      -0.19594550      0.05152022
Cr     -0.00000000      -3.42439146      -2.42141043      -0.06597455      -0.08263949      -0.18381075
0     -0.00000000      -1.71219573      -1.21070521      0.08459510      0.06942295      0.05060670
Cr     1.48280500      -2.56829360      0.00000000      0.09507719      -0.15216872      0.02774256
0      1.48280500      -0.85609787      1.21070521      -0.11916900      0.20568841      0.06652293
```

METHODOLOGY – MODEL TRAINING & ARCHITECTURE

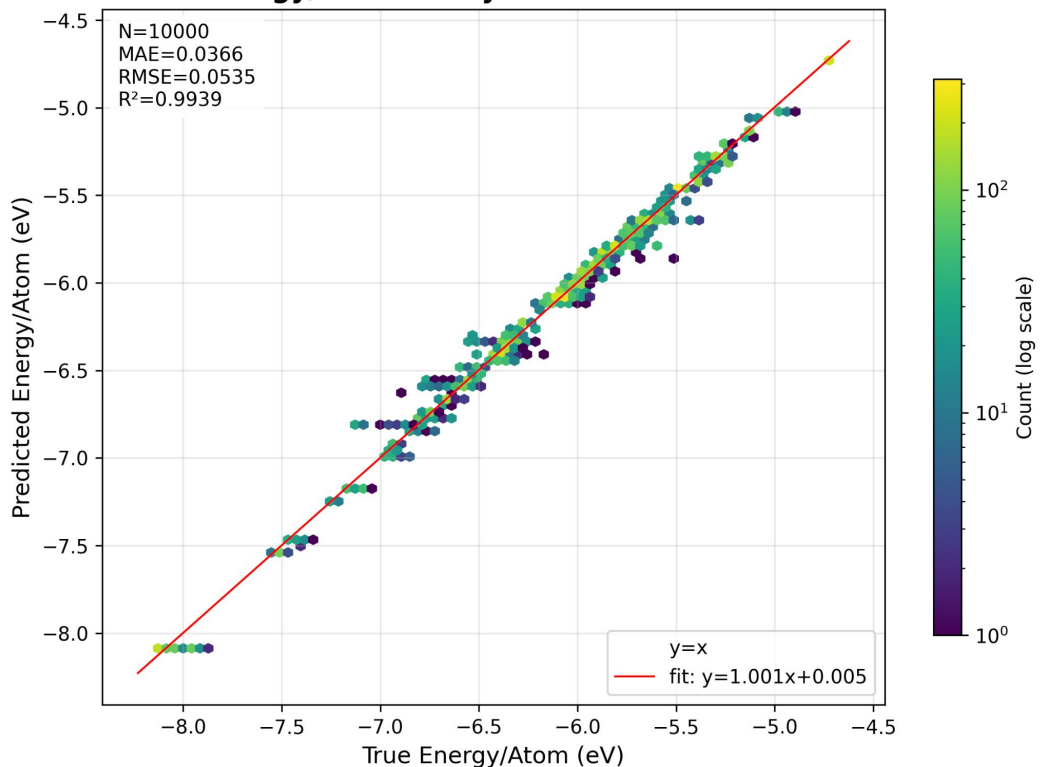
1. We pass atomic positions & species as inputs to the Allegro model to predict forces and energies. This model captures local chemical interactions within a defined cutoff radius (7.0 Å).
2. During training, we optimized the network weights by minimizing the Huber loss between the model predictions & DFT reference values.
3. We trained the model using a configuration file specifying key parameters such as the cutoff radius, chemical species, network depth and feature sizes, loss functions, optimizer settings, and training hyperparameters.

The left figure illustrates a representative subset for the number of configuration parameters used in defining the Allegro training setup.



TRAINING RESULTS

Energy/Atom Parity: Predicted vs True



ENERGY PREDICTION STATISTICS

Dataset Size:	10000 structures
Energy Range (DFT):	[-80.547, -9.449] eV
Energy Range (Predicted):	[-80.872, -9.462] eV
Mean Energy (DFT):	-42.1770 eV
Mean Energy (Predicted):	-42.1586 eV
Std Dev (DFT):	10.3671 eV
Std Dev (Predicted):	10.3489 eV

FORCE MAGNITUDE STATISTICS (Per-Atom)

Total Atoms:	69,042 atoms
Force Range (DFT):	[0.0001, 3.8903] eV/Å
Force Range (Predicted):	[0.0000, 0.0176] eV/Å
Mean Force Magnitude (DFT):	0.0879 eV/Å
Mean Force Magnitude (Predicted):	0.0025 eV/Å
RMS Force (DFT):	0.1589 eV/Å
RMS Force (Predicted):	0.0040 eV/Å
Median Force (DFT):	0.0399 eV/Å
Median Force (Predicted):	0.0013 eV/Å

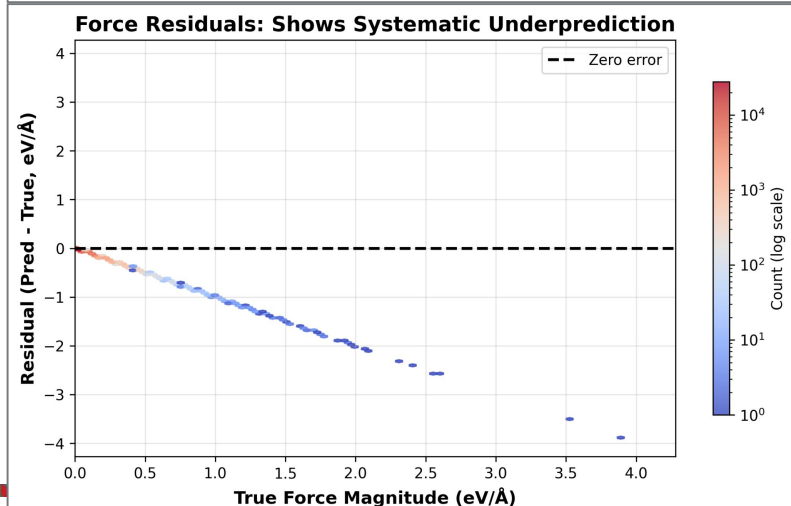
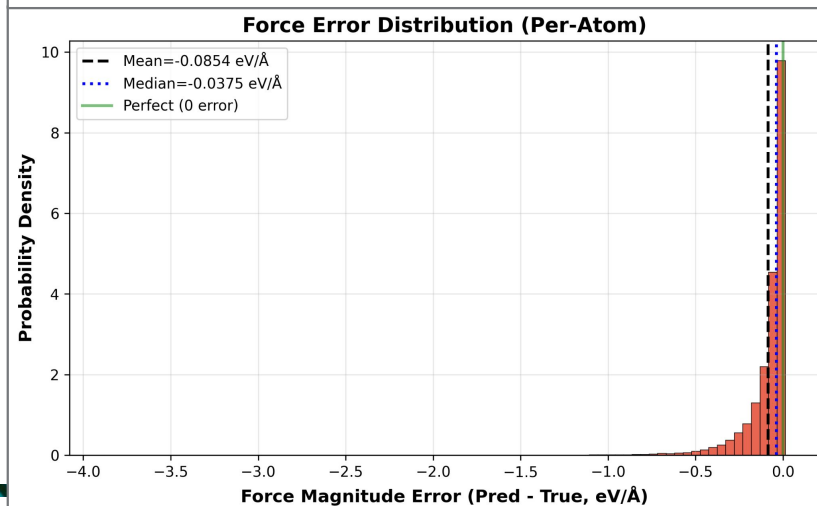
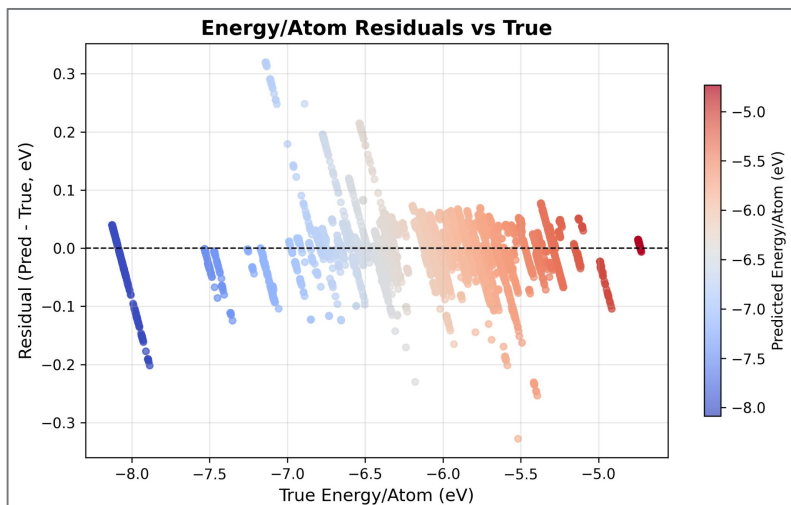
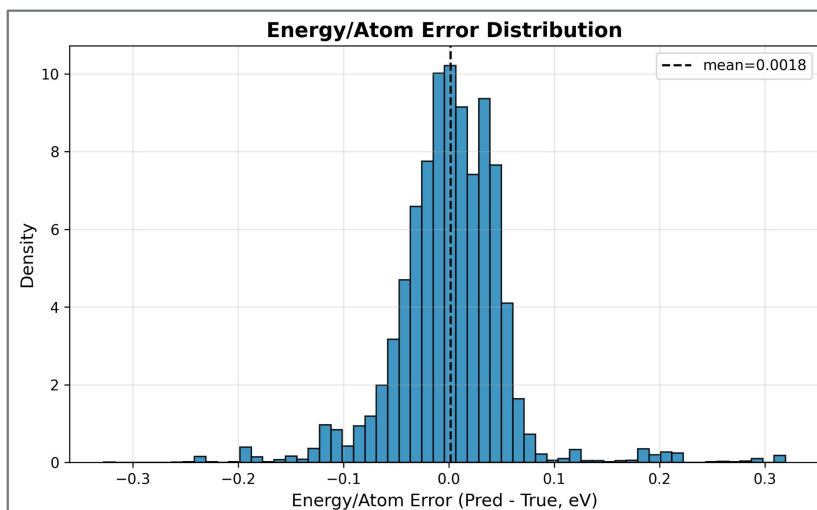
Error Metrics:

Mean Absolute Error (MAE):	0.0855 eV/Å
Root Mean Squared Error (RMSE):	0.1574 eV/Å
Max Absolute Error:	3.8838 eV/Å
Mean Error (Bias):	-0.0854 eV/Å
Median Absolute Error:	0.0375 eV/Å
R ² Score:	-0.4158

Force Distribution Analysis:

Low forces (< 0.05 eV/Å):	38,803 atoms (56.2%)
Medium forces (0.05-0.2 eV/Å):	22,190 atoms (32.1%)
High forces (> 0.2 eV/Å):	8,049 atoms (11.7%)

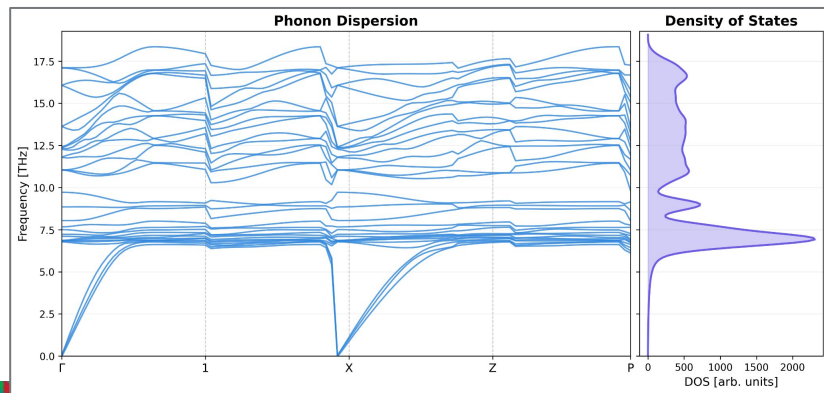
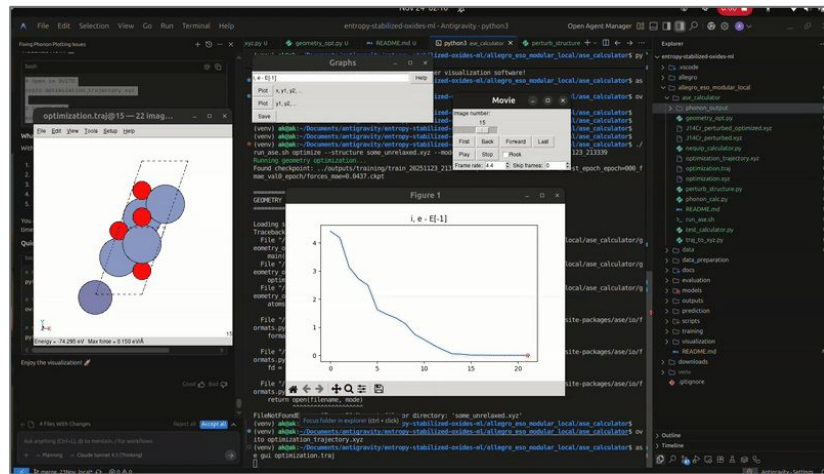
TRAINING RESULTS



METHODOLOGY – ATOMISTIC SIMULATIONS WITH ASE¹⁴

After validating the trained model for performance check, we integrate it as a calculator within the Atomic Simulation Environment (ASE). Our ASE module is designed to utilize the trained checkpoint of the model for efficient large-scale simulations like:

- **Geometry Optimization** – In this, the crystal structure is relaxed by minimizing the forces predicted by the MLIP. This process determines the equilibrium lattice parameters of the distorted HEO lattice.
- **Phonon Calculations** – Vibrational modes are computed using the Finite Displacement Method, where atoms are slightly displaced to calculate restoring forces.



CONCLUSION & NEXT STEPS

- The current work demonstrates the effectiveness of machine learning potentials in accelerating the study of complex entropy-stabilized oxides.
- By training an Allegro potential on DFT data, we established a workflow that predicts HEO properties with high computational efficiency.
- We also showed that integrating the trained Allegro model with ASE enables seamless application of the potential within standard materials science workflows.
- As next steps, this baseline setup can be extended into a fully working end-to-end pipeline. Future work includes training the Allegro model using the allotted Bridges-2 GPU resources and expanding the study toward detailed chemical characterization of HEO systems.